

Betopia Group Presents DUET Inter University Programming Contest (IUPC) 2025

<https://toph.co/c/duet-inter-university-iupc-2025>



Schedule

The contest will run for **5h0m0s**.

Rules

This contest is formatted as per the official rules of ICPC Regional Programming Contests.

You can use C++17 GCC 13.2, C++20 Clang 16.0, C++20 GCC 13.2, C++23 GCC 13.2, C17 GCC 13.2, C23 GCC 13.2, Java 1.8, Kotlin 1.9, Kotlin 2.0, PyPy 7.3 (3.10), Python 3.12, and C11 GCC 13.2 in this contest.

Be fair, be honest. Plagiarism will result in disqualification. Judges' decisions will be final.

Notes

There are 10 challenges in this contest.

Please make sure this booklet contains all of the pages.

If you find any discrepancies between the printed copy and the problem statements in Toph Arena, please rely on the latter.

A. Pair Pressure

Please refer to the printed problem set to see the statement.

Example

Input	Output
3 2 998244353 3 973733479 4 998244353	8 150 4944

B. The Last Bit of Us

Please refer to the printed problem set to see the statement.

Example

Input	Output
2 8 0 0 1 1 0 1 0 0 2 8 7 5 1 8 5 1 6 7 3 5 4 3 6 0 0 0 1 0 1 3 6 4 2 2 5 6 1 1 2	-1 3

C. Triangle Trap

Please refer to the printed problem set to see the statement.

Example

Input	Output
2 4 1 1 1 5 5 1 5 5 7 1 2 1 7 2 1 2 8 3 4 5 2 7 1	-1 5 1 7 4

D. An Interesting Problem

Please refer to the printed problem set to see the statement.

Example

Input	Output
2 2 ab 7 abababc	YES NO

E. Polynomial K Paths

Please refer to the printed problem set to see the statement.

Example

Input	Output
1 6 7 4 3 0 1 0 2 1 2 1 3 1 4 1 5 2 6 3 6 4 6	84

F. Distinct of Distincts

Please refer to the printed problem set to see the statement.

Example

Input	Output
2 2 2 1 4	2 2 5 5 5 2 6 2 3 6

G. To Infinity and Beyond

Please refer to the printed problem set to see the statement.

Example

Input	Output
2 3 -1 1 1 -1 1 2 4 -1 1 1 2 -1 1 2 1	2 1

H. Litmus Test

Please refer to the printed problem set to see the statement.

Example

Input	Output
6	5
5	1
AAAAA	2
2	2
AB	1
1	1
S	
4	
SSBB	
4	
BASS	
5	
AAAAB	

I. XOR This OR That

Please refer to the printed problem set to see the statement.

Example

Input	Output
3	161
3	26
12 23 11	6
5	
8 17 9 12 19	
2	
3 2	

J. LCM Factorization

Please refer to the printed problem set to see the statement.

Example

Input	Output
3 4 2 2 1 3 4 3 3 2 2 2 1 1 1	19 2 0